

Table S: **Interaction quality in VHSim.** Unaggregated scores for layout review.

Assistant	Overall	Factual grounding	Situational relevance	Actionable guidance	User intent uptake	Guidance conciseness
<i>Frontier API models</i>						
GPT-6 Astra	34	77	84	57	79	73
Gemini 3.1 Pro	31	71	78	61	68	69
Gemini 3.8 Flash	46	83	83	68	63	79
<i>Real-time streaming API models</i>						
Gemini 3.8 Live (polled)	42	90	76	62	52	81
GPT-Realtime 2.1 (polled)	37	82	88	67	77	54
<i>Open-weight non-streaming VLMs</i>						
Qwen3.8-27B	29	69	79	63	67	59
Molmo2-8B	17	72	56	42	20	42
InternVideo3-8B	18	64	62	45	24	44
GLM-4.6V-Flash	43	91	68	55	46	77
Kimi-VL-A3B-Thinking	12	56	60	42	37	31
Muse Glimmer	26	70	85	63	76	45
<i>Open-weight streaming VLMs</i>						
MiniCPM-o 4.5	16	62	59	35	39	29
Qwen3-Omni-30B	30	76	83	61	60	79
Proact-VL	16	70	30	22	21	26
StreamingVLM	23	77	34	29	29	20

Scores are on a 0–100 scale (run 1). Overall: full 16-item CookSim rubric with mandatory-failure gating, not the mean of the five displayed dimensions. Dimension scores are averaged within episodes and then across episodes. Bold: highest score per column.